Introduction to experimental design: Research methods in linguistics

Details:

Instructor: Joseph Casillas

Class: M / W 2:00-3:15 ML 450

Office: 209 Modern Languages

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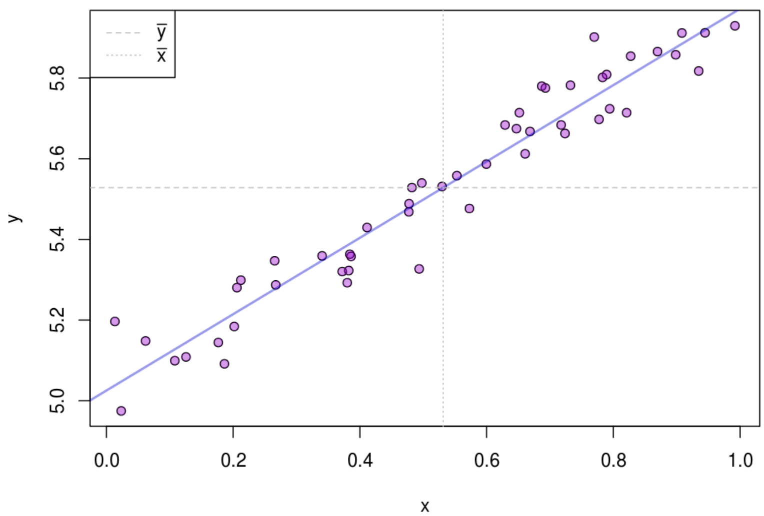
Office hours: M 12-2, W 8-10 and by appointment

Evaluation

- Participation 10%
- Homework 40%
- Quizzes 20%
- Exams 30%

Grading scale

- A = 90 - 100%
- B = 80 - 89%
- C = 70 - 79%
- D = 60 - 69%
- E = 0 - 59%



Overview

The emphasis of this course is on developing a solid understanding of the experimental paradigms and statistical procedures used in the field of linguistics. This includes, but is not limited to: sociolinguistics, phonetics, psycholinguistics, syntax, and corpus linguistics. The course is not designed to give a detailed overview of each area, thus it is assumed that the student enters the class with basic knowledge of the field of linguistics. A secondary emphasis is placed on building a solid foundation in programming in R, as well as learning the most common tools at the disposal of todays data scientist (i.e. GitHub, Cross-Validated, Knitr, etc.). Finally, this course aims to prepare the student for independent research. To this end, there will also be an emphasis on learning to find relevant information in areas that are unknown to the student. In-class participation and at-home exploration are fundamental for the student to be successful. It is recommended that personal computers be brought to each class meeting, if possible.

Materials

- Johnson, K. (2011). Quantitative Methods in Linguistics. John Wiley & Sons.
- Understanding Regression Analysis (QASS #57), SAGE Publications (ISBN: 0803927584)
- Multiple Regression in Practice (QASS #50), SAGE Publications (ISBN: 0803920547)
- GitHub account (free)
- RStudio (free)

Participation and attendance

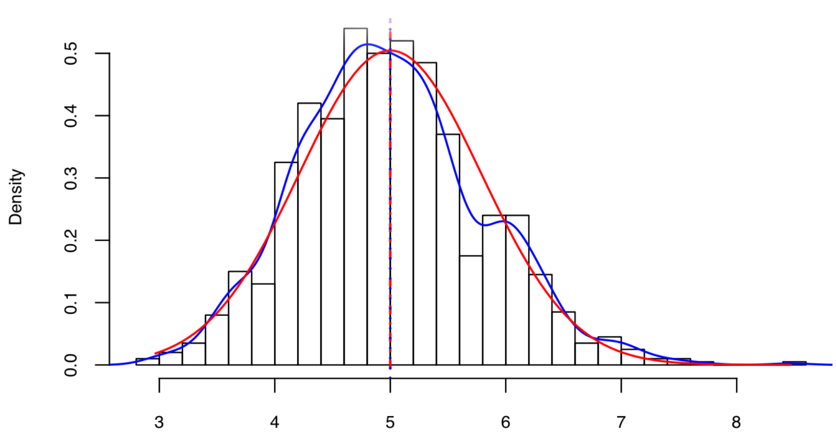
Class participation is crucial to the students success in this class. Three unexcused absences are permitted with a no-questions-asked policy; however, the student should do his/her best to communicate with the professor if/when they will be missing class. After the 3rd absence, the students' grade will be lowered by 5% each day.

Homework

Homework must be handed in on the assigned due dates. Late work is accepted with a penalty of 10% off the final grade per day late. Incomplete assignments are also accepted. In these cases partial credit is given as a function of the amount of work completed. Homework must be handed in through the course website. Programming assignments should be reproducible and in the correct format (i.e. .R, .csv, .Rmd, etc.). There will be 1-2 homework assignments per week, and they will increase in difficulty throughout the course. These assignments serve as preparation for the exams and quizzes and will give the student a baseline measure of how well they have understood the material.

Exams

There will be two exams: a mid-term and a final. The student is expected to demonstrate command of the material studied up to the week before the exam. The format is both written and electronic. The student will be given a dataset, and asked to perform a specific analysis. It is necessary that he/she explain and justify every decision made, and that the final document be reproducible. Practice exams and answer keys are available on the course website.



Policies against threatening behavior

Any behavior that is seen as disruptive to the classroom environment or is threatening in any way will not be tolerated. Please see the Code of Academic Integrity and the Code of Student Conduct online at:
http://dos.web.arizona.edu/uapolicies/

Students with disabilities

Students with disabilities must be registered with the Disability Resource Center, and must submit appropriate documentation to the instructor when requesting reasonable accommodations for class activities.
http://drc.arizona.edu/instructor/syllabusstatement

Code of Academic Integrity

The instructor will initiate an academic integrity case against students suspected of cheating, plagiarizing, or aiding others in dishonest academic behavior. Students are responsible for reading and understanding the Code of Academic Integrity.

Any work which is submitted for a grade must be 100 % the student's own work. If you are not sure when it is appropriate to seek help, please see your instructor.

Schedule

Week	Topic	Reading	Assignments
1	Introduction: Coding, R, and sampling distributions	Johnson Ch. 1	- Github account - Download RStudio
2	Introduction: Coding, R, and sampling distributions	Johnson Ch. 2	- Reproducible documents in Rmarkdown - Introduction to scripting in R
3	Phonetics: experimental designs and multiple regression	Johnson Ch. 3, QASS #57	- Comparing means - Least squares estimation - HW 1
4	Phonetics: experimental design and multiple regression	Johnson Ch. 3, QASS #57	- Quiz 1 - Introduction to Knitr - More uses of Github
5	Psycholinguistics: ANOVA	Johnson Ch. 4	- HW 2 - Presentations in Rmd
6	Psycholinguistics: Repeated measures, Interactions	Johnson Ch. 4	- Quiz 2
7	Sociolinguistics: Probabilities, logistic regression	Johnson Ch. 5	- Varbrul analysis - Contingency tables
8	Sociolinguistics: Probabilities, logistic regression	Johnson Ch. 5	- Varbrul R comparison - HW 3 - Quiz 3
9	Mid-term		
10	Phonetics II: Multiple regression	QASS #50	- Least squares estimation - Residuals
11	Phonetics II: Multiple regression	QASS #50	- Model comparison - Model assessment
12	Psycholinguistics II: Design	Handouts	- HW 4
13	Psycholinguistics II: Design	Handouts	- Quiz 4
14	Syntax: Linear mixed effects	Johnson Ch. 7	- Variability - Random slopes/int
15	Review		
16	Final exam		

Resources

Downloads

- R
- RStudio
- SublimeText3
- LaTeX (windows) (mac)

Information

- How to ask a question (on stackoverflow)
- LaTeX for linguists
- R cheatsheet
- RMarkdown cheatsheet
- Bivariate regression
- Power/Sample size calculator
- Using PsychoPy2 for linguistic research
- Logistic regression 101